$$P_A(X) = \det(A - X I_n) = \begin{vmatrix} a_{11}-X & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22}-X & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn}-X \end{vmatrix}$$

$A = (a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(k)$

$P_A \in k[X]$ $\quad$ $\operatorname{grad} P_A = n$

$$P_A(X) = (-1)^n X^n + (-1)^{n-1} \operatorname{tr}(A) X^{n-1} + \dots + \det(A)$$

$P_A(0) = \det(A)$

$\lambda \in \underline{\sigma(A)} \iff P_A(\lambda) = 0$

spectrul matricei $A$

Obs. 1) $\lambda = 0 \in \sigma(A) \iff P_A(0) = 0 \iff \det(A) = 0 \iff A$ neinversabilă

2) Thm. Hamilton-Cayley: $\forall A \in \mathcal{M}_n(k)$, $P_A(A) = O_n$

exemplu $n = 2$ $\quad P_A(X) = X^2 - \operatorname{tr}(A) X + \det(A)$

$$P_A(A) = O_2 \iff A^2 - tr(A)A + \det(A)\cdot \frac{I}{2} = O_2$$

$n = 3$ $\qquad A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{pmatrix}$ $\longrightarrow$ teme: să se aducă la forma diag. $A = P \cdot \Delta \cdot P^{-1}$ $P_2$ -- $D_2$ --

$P_A(X) = ?$

$$P_A(X) = \det(A - XI_3) = \begin{vmatrix} 1-X & -1 & 2 \\ 0 & 2-X & -4 \\ 0 & 0 & 3-X \end{vmatrix} =$$

$$= \underbrace{(1-X)^1(2-X)^1(3-X)^1} = -X^3 + 6X^2 - 11X + 6$$

$$-A^3 + 6A^2 - 11A + 6I_3 = O_3$$

$$I_3 = \frac{1}{6}\left(A^3 - 6A^2 + 11A\right) \Rightarrow$$

$$I_3 = A \cdot \underbrace{\frac{1}{6}\left(A^2 - 6A + 11I_3\right)}_{A^{-1}}$$

$$A^3 = 6A^2 - 11A + 6I_3 \in \mathrm{Sp}\{I_3, A, A^2\}$$

$$A^4 = 6A^3 - 11A^2 + 6A$$

$$= 6(6A^2 - 11A + 6I_3) - 11A^2 + 6A$$

$$A^4 \in \mathrm{Sp}\{I_3, A, A^2\}$$

$$A^k \in \mathrm{Sp}\{I_3, A, A^2\} \quad \forall\, k \in \mathbb{Z}$$

(dc $A$ inversabilă
altfel $\forall\, k \in \mathbb{N}$)

---

$K = \mathbb{R} \subseteq \mathbb{C} \Rightarrow P_A(X) \in \mathbb{R}[X] \Rightarrow \lambda, \bar{\lambda}$

$A \in M_n(\mathbb{R}) \subseteq M_n(\mathbb{C}) \qquad V_\lambda \subseteq \mathbb{C}^n$

$\lambda \in \mathbb{C}$ valoare proprie $\quad v \in V_\lambda$

$Av = \lambda v \Rightarrow A\bar{v} = \overline{\lambda v} = \bar{\lambda} \cdot \bar{v}$

exemplu. $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R}) \subseteq M_2(\mathbb{C})$

$P_A(X) = X^2 + 1 \begin{cases} \lambda_1 = i \\ \lambda_2 = -i \end{cases}$

- $V_{\lambda_1} = \{ v \in \mathbb{C}^2 \mid Av = \lambda_1 v \}$

$v = \begin{pmatrix} z \\ w \end{pmatrix}$ $\quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = i \begin{pmatrix} z \\ w \end{pmatrix}$

$\Rightarrow \begin{cases} -w = iz \\ z = iw \end{cases}$

$V_{\lambda_1} = \left\{ \begin{pmatrix} iw \\ w \end{pmatrix} \mid w \in \mathbb{C} \right\} = Sp\left\{ \begin{pmatrix} i \\ 1 \end{pmatrix} \right\}$

$\Rightarrow$ - $V_{\lambda_2} = Sp\left\{ \begin{pmatrix} -i \\ 1 \end{pmatrix} \right\}$

<u>obs.</u> thm. fundamentală a algebrei

$\forall P \in \mathbb{C}[X]$ $\exists$ cel puțin o rădăcină $\lambda \in \mathbb{C}$ pt $P(X)$

(Corpul $\mathbb{C}$ este algebric închis)

$\mathbb{Q}$ $\quad X^2 - 2$ $\quad \mathbb{Z}_p$ ?

$\mathbb{R}$ $\quad x^2+1$

$\cap\!\!|$

$\mathbb{C}$ $\quad \ldots\ldots$

$\mathbb{Z}_2 \leftarrow (x^2+x+1)$

$\cap\!\!|$

corp $k$ 4 elemente

$\text{card}(k) = 4$

$k = \{a_1, a_2, a_3, a_4\}$

$k \times k \xrightarrow{\omega} k$ (16, 4)

$4 \cdot 4 \cdot \ldots \cdot 4 = 4^{16}$ (16 factori)

$\omega(a_1, a_1) = \ldots = 4$

$\omega(a_1, a_2) = \ldots = 4$

$\ldots\ldots$

$\omega(a_4, a_4) = \ldots = 4$

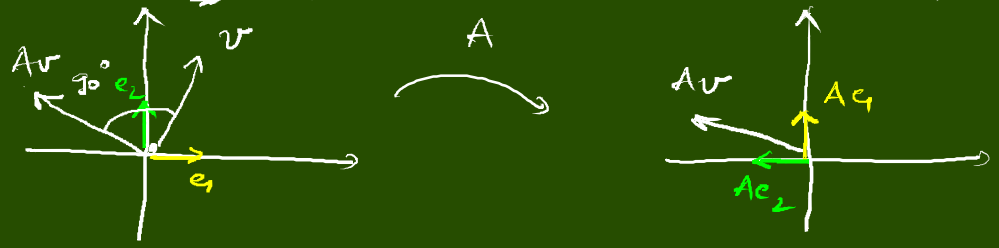
$C_{4^{16}}^{2} + C_{4^{16}}^{1}$

$\ldots\ldots$

corp

---

$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ notație cu $\frac{\pi}{2}$ (trigon)



$$Ae_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e_2$$

$$Ae_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} = -e_1$$

$v \in \mathbb{R}^2$ vect. pr. $\Leftrightarrow v \neq 0$ $\underline{Av = \lambda v}$

$Av$, $v$ colinéaires

---

**Prop.** $\lambda_1, \dots \lambda_p \in \sigma(A)$ $\quad A \in \mathcal{M}_n(k)$

$\lambda_i \neq \lambda_j$ , $i \neq j$

$v_1 \in V_{\lambda_1}$ $\quad Av_1 = \lambda_1 v_1$ $\quad v_1 \neq 0$

$\vdots$

$v_p \in V_{\lambda_p}$ $\quad Av_p = \lambda_p v_p$ $\quad v_p \neq 0$

$\Rightarrow v_1 \dots v_p$ lin. indép.

**Dem.** $p = 1$ $\quad v_1 \neq 0 \Rightarrow v_1$ lin. indép.

$p \Rightarrow p+1$ $\quad$ Soit $v_1, \dots v_{p+1}$ vect. pr. assoc.

val. pr. distincte $\lambda_1 \dots \lambda_{p+1}$

Fie $\alpha_1 \dots \alpha_{p+1} \in K$ a.i. $\alpha_1 v_1 + \dots + \alpha_{p+1} v_{p+1} = 0 \mid A$

$\Rightarrow \alpha_1 A v_1 + \dots + \alpha_{p+1} A v_{p+1} = 0$

$$\begin{cases} \alpha_1 \lambda_1 v_1 + \dots + \alpha_{p+1} \lambda_{p+1} v_{p+1} = 0 \\ \alpha_1 v_1 + \dots + \alpha_{p+1} v_{p+1} = 0 \mid \lambda_{p+1} \\ \alpha_1 \lambda_{p+1} v_1 + \dots + \alpha_{p+1} \lambda_{p+1} v_{p+1} = 0 \end{cases}$$

$\Rightarrow \alpha_1 \underbrace{(\lambda_1 - \lambda_{p+1})}_{\neq 0} v_1 + \dots + \alpha_p \underbrace{(\lambda_p - \lambda_{p+1})}_{\neq 0} v_p = 0$

$v_1 \dots v_p$ vect. pr. corresp. val. pr. $\neq$ $\lambda_1 \dots \lambda_p \Rightarrow$ lin. indep. $\Rightarrow$

$\Rightarrow \alpha_1 \underbrace{(\lambda_1 - \lambda_{p+1})}_{\neq 0} = \dots = \alpha_p \underbrace{(\lambda_p - \lambda_{p+1})}_{\neq 0} = 0$

$\Rightarrow \alpha_1 = \dots = \alpha_p = 0 \Rightarrow \alpha_{p+1} = 0$

Consec. $\lambda_1 \neq \lambda_2$ val. pr.

dacă există $v \neq 0$ $\quad v \in V_{\lambda_1} \cap V_{\lambda_2} = \{0\}$

$\Rightarrow$ $v$, $v$ liniar indep., fals

$(v - v = 0)$ $\Rightarrow$ $V_{\lambda_1} \cap V_{\lambda_2} = \{0\}$

$V_{\lambda_1} \oplus V_{\lambda_2}$

În general, $\lambda_1 \dots \lambda_p$ val. pr. distincte $\lambda_i \neq \lambda_j$ $\Rightarrow$

$\Rightarrow$ sunt subspații vectoriale proprii $V_{\lambda_1}, \dots, V_{\lambda_p} \subseteq K^n$

este directă: $V_{\lambda_1} \oplus V_{\lambda_2} \oplus \dots \oplus V_{\lambda_p}$

$\mathcal{B}_1 \cup \mathcal{B}_2 \cup \dots \cup \mathcal{B}_p$ mulțime de vectori liniar indep.

$\leq n$

$\mathbb{R}^3$ $\quad \mathcal{B}_1 = \{e_1, e_2\}$ $\quad \mathcal{B}_2 = \left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, e_3\right\}$

$B_1 \cup B_2$ unione di vettori linearmente dipendenti

**esempio.** $A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$ $\sigma(A)$

$\forall \lambda \in \sigma(A)$

$a(\lambda) = ?$

$g(\lambda) = ?$

$$P_A(x) = \begin{vmatrix} -2-x & 1 & 1 \\ 1 & -2-x & 1 \\ 1 & 1 & -2-x \end{vmatrix} =$$

$$= \begin{vmatrix} -x & 1 & 1 \\ -x & -2-x & 1 \\ -x & 1 & -2-x \end{vmatrix} =$$

$$= (-x) \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2-x & 1 \\ 1 & 1 & -2-x \end{vmatrix} =$$

$$= (-x) \begin{vmatrix} 1 & 1 & 1 \\ 0 & -3-x & 0 \end{vmatrix}$$

$$\begin{vmatrix} -x & 1 & 1 \\ 1 & -x & 1 \\ 1 & 1 & -x \end{vmatrix} \quad \begin{matrix} \\ 0 & 0 & -3-x \end{matrix} \quad \begin{matrix} \\ 1 \end{matrix}$$

$$= (-x)(-3-x)^2 = (-1)^3 \cdot (x-0)^1 (x+3)^2$$

$\lambda_1 = 0$ ↳ $a(\lambda_1) = 1$

$\lambda_2 = -3$ ↳ $a(\lambda_2) = 2$

$$P_A(x) = (-1)^n (x-\lambda_1)^{n_1} \cdots (x-\lambda_p)^{n_p} \qquad \lambda_i \neq \lambda_j,\ i \neq j$$

$\lambda_1$ rădăcină cu ordinul de multiplicitate $n_1$ ($a(\lambda_1) = n_1$)

$\boxed{\lambda_1 = 0}$ $\quad V_{\lambda_1} = \{ v \in \mathbb{R}^3 \mid Av = \lambda_1 v \}$

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$v$

$$\begin{cases} -2x + y + z = 0 \\ x - 2y + z = 0 \\ x + y - 2z = 0 \end{cases} \quad -3x + 3y = 0,\ x = y \Rightarrow z = x = y$$

$\Rightarrow V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$

$\dim_{\mathbb{R}} V_{\lambda_1} = 1 \qquad g(\lambda_1) = 1$

$\lambda_2 = -3 \qquad V_{\lambda_2} = \dots$

$$\Rightarrow \begin{cases} -2x + y + z = -3x \\ x - 2y + z = -3y \\ x + y - 2z = -3z \end{cases}$$

$\Rightarrow \quad x + y + z = 0 \qquad z = -x - y$

$V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} \mid x, y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$

$\dim_{\mathbb{R}} V_{\lambda_2} = 2 \qquad g(\lambda_2) = 2$

$\Rightarrow$ în acest exemplu $a(\lambda_1) = g(\lambda_1)$

$a(\lambda_2) = g(\lambda_2)$

<u>Temă</u>: $A = \begin{pmatrix} -2 & 1 & -14 \\ 0 & -2 & -5 \\ 0 & 0 & 3 \end{pmatrix}$

$\Rightarrow \quad \lambda_1 = -2 \quad a(\lambda_1) = 2 \geq g(\lambda_1) = 1$

$\lambda_2 = 3 \quad a(\lambda_2) = 1 \geq g(\lambda_2) = 1$

$\Rightarrow \quad \forall \lambda \in \sigma(A)$

$$1 \leq \underbrace{g(\lambda) \leq a(\lambda)}_{1} \leq n$$

---

$A, B \in M_n(k)$ $A, B$ matrice simile

dc. $A = P.B.P^{-1} \Rightarrow P_A(x) = P_B(x)$

---

esempio $A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$

$\left( \begin{array}{l} \lambda_1 = 0 \quad a(\lambda_1) = g(\lambda_1) = 1 \\ \lambda_2 = -3 \quad a(\lambda_2) = g(\lambda_2) = 2 \end{array} \right| \Rightarrow$

$\Rightarrow$ $A$ diagonalizzabile. $A = P.D.P^{-1}$

Algoritm de diagonalizare pt $A \in \mathcal{M}_n(K)$

1) $P_A(X) = \det(A - X I_n)$

2) rădăcinile $P_A(X)$ $\lambda$ ... $\lambda$ multiplicități ac. $a(\lambda)$

3) $\forall \lambda \longrightarrow V_\lambda$ → bază $v_1^\lambda \dots v_k^\lambda$
→ dim $g(\lambda)$

4) decidă: $\forall \lambda$ $g(\lambda) = a(\lambda) \longrightarrow A$ diag. ✓

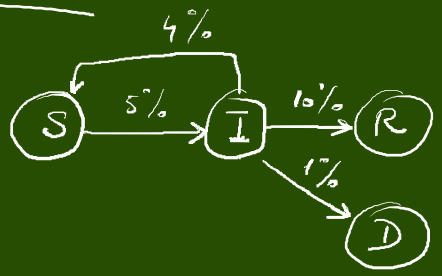
5) $A = P \cdot D \cdot P^{-1}$

$P = (v_1 | \dots | v_n)$

reuniunea bazelor subsp. proprii

$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

Example. 1) SIR




$$A = \begin{array}{c} \\ S \\ I \\ R \\ D \end{array} \begin{array}{c} \begin{array}{cccc} S & I & R & D \end{array} \\ \begin{pmatrix} 0{,}95 & 0{,}04 & 0 & 0 \\ 0{,}05 & 0{,}85 & 0 & 0 \\ 0 & 0{,}10 & 1 & 0 \\ 0 & 0{,}01 & 0 & 1 \end{pmatrix} \end{array}$$

$$v(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$v(1) = A\,v(0) = \begin{pmatrix} 0{,}95 \\ 0{,}05 \\ 0 \\ 0 \end{pmatrix}$$

$$v(2) = A\,v(1) = A^2 v(0)$$

$$v(n) = A^n v(0)$$

$$A = P \cdot D \cdot P^{-1}$$

$$A^n = (P \cdot D \cdot P^{-1})^n$$

$$= P \cdot D \cdot \cancel{P^{-1}} \cancel{P} \cdot D \cdot \cancel{P^{-1}} \cdots \cancel{P} D P^{-1}$$

$$A^n = P D^n P^{-1}$$

$$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \longrightarrow D^n = \begin{pmatrix} \lambda_1^n & & \\ & \ddots & \\ & & \lambda_n^n \end{pmatrix}$$